

REVIEW PAPER

European beech (*Fagus sylvatica* L.): A promising candidate for future forest ecosystems in Central Europe amid climate change

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Abstract

On the one hand, the European beech (*Fagus sylvatica* L.) is the tree of the future due to ongoing climate changes, on the other hand, there are questions about its expansion and economic use as a replacement for the declining Norway spruce (*Picea abies* [L.] Karst.). This literature review examines 140 studies summarizing basic research on beech in the context of climate change. As a climax tree species, beech is becoming dominant again in parts of its original range at the middle and higher altitudes of Central Europe, following spruce. It is a shade-loving species that can thrive in various types of mixed forest stands. To cultivate beech, close-to-nature methods, shelterwood, or selection management are optimal. The occurrence of the beech seed year is influenced by factors such as precipitation, temperature, drought, and air pollution. Although beech is generally considered resistant to abiotic and biotic factors, it often needs protection against hooved game browsing in the earliest stages of development. As climate change progresses, it is essential to cultivate beech in areas rich in precipitation and nutrients where it can thrive even under more extreme conditions. In optimal conditions, beech has shown intensive regeneration in recent years, aggressively displacing other tree species. However, with insufficient precipitation and prolonged periods of drought, beech loses its vigor, production, and ability to compete. For adaptation to climate change, it is recommended to select appropriate beech provenance, promote natural regeneration, and cultivate structurally differentiated stands through positive-selection thinning from above.

Key words: close-to-nature silviculture; natural regeneration; production; ecological requirements; threats

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1. Introduction

European beech is a tree species typical of oceanic and suboceanic climates. It is susceptible to drought and late frosts (Ningre & Colin 2007; Gazol et al. 2019). Beech forests form Central Europe's most important forest ecosystems, primarily at altitudes between 100 and 900 meters above sea level (von Oheimb et al. 2005). Beech is profoundly sciophilous, i.e., tolerates permanent shade, and therefore, is suitable for shelterwood and selection silvicultural systems (Barna 2011; Barna et al. 2011). Beech stands also endure thinning well, even at an older age, because their crowns respond well to release (Štefančík 2015; Štefančík et al. 2018a). Beech

is a dominant tree species in a large spectrum of habitats, except for waterlogged soil, and is classified as a climax tree species (Slezák et al. 2016). Its representation in the Czech Republic, as well as in Central Europe, decreased significantly in the past. Its natural representation in the Czech Republic is 40.2%. Currently, it is only 9.3%, while the recommended share is 18.0% (MZE 2022). The shade-loving beech has the prerequisites for spontaneous natural regeneration in old stands with reduced stand density (Barna & Bošela 2015). However, on sites lacking the protection of the parent stand, its natural regeneration often suffers from late frosts and excessive weeds. To obtain high-quality wood, we need high-density and healthy young growth on suitable sites (Štefančík 2015).

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The ability of beech to regenerate naturally is reduced as a consequence of a significant decrease in the quantity and quality of fructification due to air pollution (Vacek 1988). Additionally, the high fructification irregularity causes problematic and unreliable natural regeneration (Vacek & Mareš 1985). When beech is reintroduced, underplanting appears to be a suitable implementation method. In clearings, it should be planted using the ecological shelter of mature stands (Podrázský et al. 2019). Currently, beech is threatened by climate change, causing higher average temperatures and more frequent long-term droughts, affecting the growth, competitiveness, and shifts in distribution ranges of this tree species (Knutzen et al. 2017; Vacek et al. 2023).

The radial increment of European beech is better adapted to climatic fluctuations at lower and middle altitudes and in nutrient-rich habitats (Aertsen et al. 2014; Vacek et al. 2019). At lower altitudes, beech is more susceptible to fluctuations such as drought or summer heat. At higher altitudes, however, beech is limited by low temperatures and spring frosts, which can significantly damage it, especially during shooting (Králíček et al. 2017; Šimůnek et al. 2019; Vančura et al. 2022). The limitation of growth by drought is also evident in the southern range of this tree species – in the Mediterranean mountains (Tegel et al. 2014; Tognetti et al. 2019), whereas in the northern range, the growth is limited by low temperatures (Dittmar et al. 2003; Dorado-Liñán et al. 2019). During the ongoing climate change, the European beech expands northwards in Europe even at lower altitudes, while contrastingly, it is the high mountain areas in the southern part of the distribution. Beech also has the potential for afforestation of agricultural land (Penuelas & Boada 2003; Bolte et al. 2010; Kramer et al. 2010; Cukor et al. 2022; Vacek et al. 2023).

Beech stands are also affected by cyclical events during climate change, which subsequently are reflected in their increment. One commonly recurring cyclical influence is the solar cycle, which has been confirmed in the tree-ring growth of beech stands across Europe (Šimůnek et al. 2020, 2021; Komitov 2021). However, the most important cycles include the temperature (7-year and 3-year cycles) and precipitation (3.3-year and 2.4-year cycles). Climatic cycles are also indirectly linked to more frequent beech stand fructification. According to a study from Sweden, fruiting has been more frequent in the last few decades, at an interval of three to four years (Drobyshchev et al. 2014). Seed years directly depend on summer temperatures, which influence fructification. Positive temperature anomalies affect the seed year one to two vegetation seasons before a massive seed year. Recently, beech stands have become more prolific, contributing to their intensified expansion in Central and Northern Europe, where they have also been shifting to higher altitudes (Saltré et al. 2013; Pavlović et al. 2019). Due to its plasticity, European beech appears to be one of the most promising tree species for tolerating higher mean annual

temperatures caused by climate change, which has had a positive effect on this species in recent years.

This literature review of 140 studies aims to assess the role, opportunities, and risks of European beech in European forestry. Specifically, the aim is to focus on a detailed review of (i) the species' description and distribution, (ii) ecological requirements, (iii) threats and diseases, (iv) habitat and stand conditions, and (v) close-to-nature silvicultural management, all in the face of ongoing climate change.

2. Description and distribution

European beech is a large deciduous tree growing 35 to 40 m tall (von Wuehlisch 2008). The root system is heart-shaped or sometimes taproot-like, strongly branched with strong lateral roots (Kodrík & Kodrík 1999). The trunk is slender and cylindrical, and may occasionally show longitudinal callus tissue, and can be twisted or transversely corrugated. It reaches a diameter of up to 1 m. The bark is smooth, grey to white-grey, rarely fissured, with various bark forms (von Wuehlisch 2008). The crown is set high and spherical, richly branched, and variable in shape, with strong branches usually angled away from the trunk (Wagner et al. 2010). The branching of beech is variable, determined by environmental – especially light – conditions and interspecific competition (Bayer et al. 2013). The crown morphology is characterized by forks, negatively influencing the distribution of wood mass (Vacek 1987). Annual shoots are reddish-brown, initially whitish-hairy, and later glabrous. Buds are zigzag, slenderly spindle-shaped, 10–25 mm long, and cinnamon brown. Scales are whitish at the tip. Leaves are short-petiolate, the blade is elliptic to ovate-elliptic, 5–7 cm long and 3–6 cm width (leaf area 12–26 cm²), entire or slightly serrated, wavy on the edge, glabrous and shiny on the face, paler on the underside, retaining long, white-silk hairs in the leaf axils and on larger veins (Hladká & Čaňová 2005). Lateral veins are in five to nine pairs. The petiole is 5–10 mm long and hairy. Stipules are narrowly lanceolate, light brown, and shiny. They drop soon. The male flowers are in long-stalked clusters in the leaf axils, the female flowers are in 2–3 clusters in a reddish calyx with hairy apophyses. Fruits are about 1 cm large, brown, glossy triangular, winged on the edges, achenes – beech nuts. They are enclosed by 2–3 in a brown, woody, spiny calyx, pedunculate about 2 cm large, opening by four flaps. Autumn leaf color ranges from yellow to reddish to dark brown (von Wuehlisch 2008). Beech tress can reach an age up to 400–550 years (Moning & Müller 2009; Trotsiuk et al. 2012).

Natural beech stands are characterized by lower stocking and high age diversity and are only affected by small-scale disintegration, preventing the large ones (Vacek et al. 2010b). Beech forests naturally develop in

cycles lasting approximately 250 years, going through all phases of their small-scale development cycle (Vacek et al. 2010a; Šimůnek et al. 2019). Beech trees form up to three-story stands, and independent development takes place over an area of 25–30 hectares, with beech trees naturally reproducing under the parent stand, benefiting its shade-tolerant growth at a younger age (Vacek et al. 1988; Vacek et al. 2010a).

Beech began to spread to Europe from its Balkan refugium in the central Danube region. Within our territory, it occurred sporadically at lower altitudes in the Atlantic period, with the most extensive distribution of beech recorded from the Epi-Atlantic to the Sub-Atlantic, where its range stabilizes and its share in forest stands is the highest (Giesecke et al. 2007). The current distribution of beech is shown in Fig. 1. Concerning climatic conditions, beech has a lower limit of occurrence in the warm climatic region with dry, slightly wet to cold winters. The upper climatic boundary is established as a moderately cool/cool mountain precinct in the cool climatic region. Considering the various natural conditions in which European beech can thrive, its occurrences are determined by the length of the growing season, which ranges from 90 to 175 days, with an average daily temperature above 10 °C (Dittmar et al. 2003).

3. Ecological requirements

Beech is indifferent to the soil and its subsoil. It grows in most soil types but does not thrive in heavy clay soils,

swamps, or sandy, dry soil. However, it has very good growing ability on nutrient-rich sandy soils with a higher water table. Beech growth is strongly influenced by the soil's nutrient content. On richer soils (especially in Ca), beech grows taller and more directly. Outside the area of the growth optimum, beech demands more from the soil. In cooler alpine areas, beech prefers warmer limestone and dolomite soils. In warmer locations, it favors clayish soil with high water content (Leuschner et al. 2006). In the Czech Republic, the best soil conditions for beech growth and development are in the Carpathian flysch zone. Deeper mineral soils with good physical structure are more suitable for beech. Through its root system, it can accumulate and distribute nutrients and water from deep layers. Among other things, beech can efficiently use soil moisture through its root system. Leaf fall from the beech tree has a positive effect on the humus layer; the leaves decompose within three years. The favorable litterfall and mitigation of acidification classify beech as a soil-improving tree species (Borůvka et al. 2005; Podrázský & Remeš 2007; Remeš 2007).

Beech does not require a high intensity of solar radiation for photosynthesis. Its minimum light requirement, expressed as a relative light minimum, is 1/60 to 1/80 (the ratio of the light intensity of the leaves in the darkest part of the canopy to that of the fully illuminated leaves). Even a low diffuse light intensity is sufficient for beech to grow and establish natural regeneration. This assumption is considered in the management and regeneration of beech stands. The light intensity also influences the development of older stands. Opening-up thinning improves the volume increment in older stands (Minotta & Pinzauti 1996).

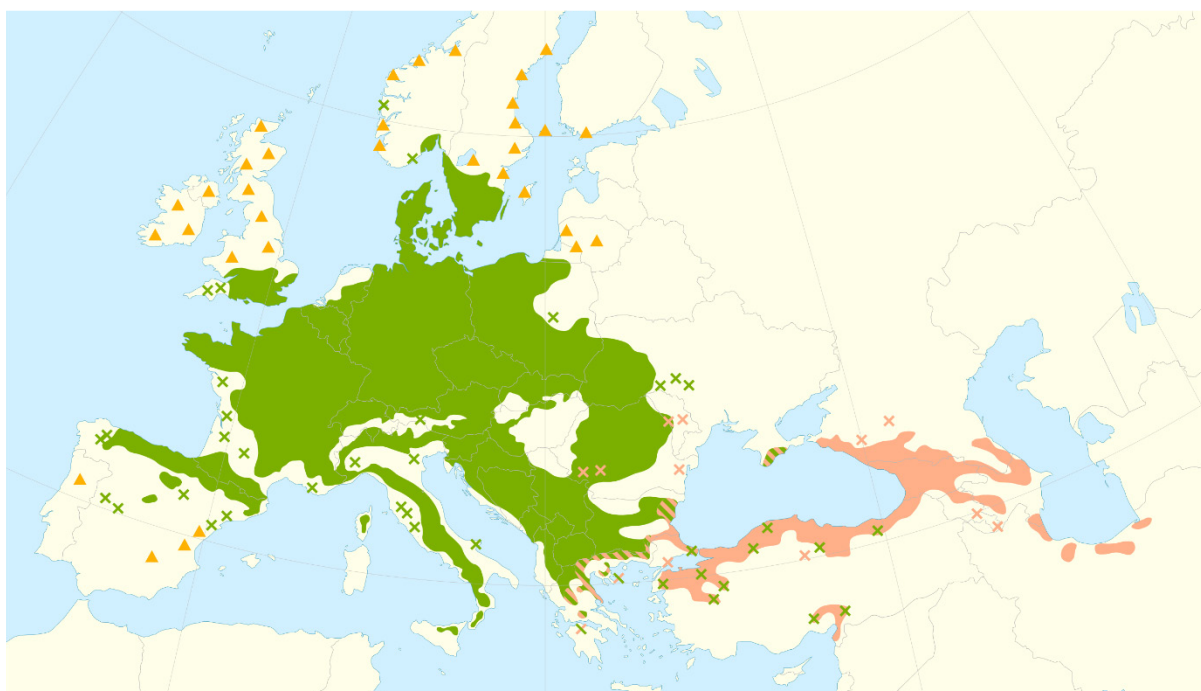


Fig. 1. Distribution area of European beech (green – European beech, pink – Oriental beech, triangle and cross – introduced and natural occurrence; Caudullo et al. 2017).

Heat, or rather, thermal extremes, affect the natural distribution limit of the species (Durrant et al. 2016). Beech favors average monthly temperatures of 15–25 °C in the growing season and around 0 °C in the coldest month of the year. Optimal beech habitats in Central Europe have an average annual temperature of approximately 10 °C. Beech needs at least 26 summer days with a maximum temperature of over 20.5 °C and can tolerate no more than 120 winter days with a maximum below 5 °C. Frost basins are the limit of beech regeneration, even at mid-altitudes. The amount of beech fructification is related to the temperature in July of the previous year and the duration of sunshine. Late frosts and short-term droughts reduce beech flowering and seed set (Latte et al. 2016). During the peak air pollution calamity years, beech reproduction was largely restricted to layering in mountainous areas. In recent years, however, the incidence of seed years has increased considerably following the air pollution abatement and climate change (Vacek & Hejzman 2012; Vacek et al. 2022).

Beech requires a humid atmosphere with precipitation well distributed throughout the year (Durrant et al. 2016). In optimal positions, precipitation is around 1,000 mm per year. In colder areas, beech only needs 500 mm per year. Compared to other tree species, beech is a medium-water-demanding species. Beech stands let a considerable amount of precipitation into the forest's ground layers, which allows natural regeneration to develop. Of the total summer rainfall, beech stands receive about 80%, whereas spruce stands receive only 59%. The precipitation release in beech stands is determined by the position of the branches and the overall morphology of the tree. Water runs down the smooth trunk towards the ground in large quantities, which is different from the coarse bark in spruce, where it tends to run down the needles and branches (Michelot et al. 2012).

4. Threats and diseases

Among abiotic factors, beech is currently threatened by long-term drought and significant temperature fluctuations (Vacek et al. 2023). It is also threatened by extreme climatic events, in particular, glaze ice and intensive snow falls which can cause branch breakage damages under the weight of the wet snow (Jarzyna 2021). The root system protects beech from strong-wind calamities (Durrant et al. 2016).

From the biotic point of view, beech is a tree species with a relatively low threat of insect pests. During the last century, beech forests have suffered from repeated outbreaks of pests such as the pale tussock (*Calliteara pudibunda* L.), the beech scale (*Cryptococcus fagisuga* Lindinger) and the lobster moth (*Stauropus fagi* L.; Gora et al. 1996; Mazzoglio et al. 2005). Among the more frequent beech insect pests belongs also gall insect *Mikiola fagi* Hartig and *Hartigiola annulipes* Hartig (Fig. 2). Nowadays, however, when the climate is changing life cycle of beech gall insect seems to be strongly affected (Pilichowski et al. 2017). Another potential threat to older beech stands in the future is the beech bark beetle (*Taphrorychus bicolor* Herbst) and the green jewel beetle (*Agrilus viridis* L.). While currently sporadic and relatively insignificant, climate change could exacerbate its impact because of increasing number of weakened trees, ideal places for mass reproduction of this two beetle species (Lakatos & Molnár 2009). Beech bark necrosis, especially *Nectria* spp., is a dangerous disease, and trees whose bark has been afflicted wither and die. Affected wood loses quality and can usually be used only for fuel, which causes significant economic losses, especially in young stands (Fig. 2). The main silviculture recommendation is to avoid the injury of the trees during the harvesting process (Mititelu et al. 2021). A severe decline and dieback of beech stands in Europe have been caused



Fig. 2. Significant harmful agents of European beech – leaf gall midge (*Mikiola fagi* Hartig) in Southern Poland (left) and bark necrosis due to *Nectria* spp. in Central Slovakia (right) (Photo: Zdeněk Vacek).

also by *Phytophthora* spp. presence, which may become a more serious problem in the future due to changing climate conditions (Corcobado et al. 2020). Old beech trees can also develop false heartwood, which is caused by a lack of calcium in the soil, dampening of the inner part of the trunk, or by frost. False heartwood does not have the typical physical and mechanical properties of wood without heartwood and is often subject to rot which reduces stability and timber value (Durrant et al. 2016).

The earliest stage of beech growth is the most critical period. In recent years, cloven-hoofed game appears to be one of the most significant limiting factors for natural regeneration (Vacek 2017). The majority of fallen beech nuts do not germinate, as they are consumed by wildlife during autumn and winter. This is particularly relevant for sites with high population densities of wild boar (*Sus scrofa* L.), whose reproduction depends on the availability of food, i.e., seed years of fruit-bearing trees, including beech (Frauendorf et al. 2016; Drimaj et al. 2020). Wild boar populations across Europe are gradually increasing (Massei et al. 2015), significantly reducing the number of seeds available for massive natural regeneration, which would subsequently resist further negative impacts as it grows. The wild boar population is unlikely to decrease through hunting management, which is ineffective in the long term, but through the spread of African swine fever throughout Central Europe (Cukor et al. 2020, 2021). Beech nuts also provide food for pigeons (*Columba palumbus* L.), finches (*Fringilla coelebs* L.), yellow-necked mice (*Apodemus flavicollis* Melchior), bank voles (*Clethrionomys glareolus* Schreber), and other wildlife species. Beech nuts are threatened by fungi (*Phytophthora cactorum* [Leb. and Cohn] Schröeter, etc.) and other fungal diseases (*Rhizoctonia solani* Kühn) (Procházková 2009). If seedlings develop, they are often very intensively damaged by cloven-hoofed animals from browsing on the terminal and lateral shoots (Clasen et al. 2011; Hejle et al. 2016). Damaged seedlings have reduced vigor and increment and become dwarfed individuals of bonsai habitus. Protection of seedlings and entire cultures is implemented by economically costly site fencing, repellents, or individual protection (Durrant et al. 2016) as the numbers of cloven-hoofed game increase (Valente et al. 2020; Carpio et al. 2021).

5. Site and stand conditions under climate change

Severe and frequent droughts are expected to become a serious risk for forest ecosystems under climate change (Kolström et al. 2011). Changing climatic conditions are causing changes in forest structure and silvicultural strategies (Fürst et al. 2007; Nabuurs et al. 2007). It is evident that coniferous monocultures, prevalent in the current forest management in Central Europe, will not be

sustainable in the future (Seidl et al. 2011). A substantial increase in air temperature accompanied by a concomitant decrease in precipitation during the growing season in Central and Southern Europe will increase the likelihood of lengthy and intense summer droughts (EEA 2004). For these reasons, mixed stands should only be established in suitable habitats (Lindner et al. 2014). Coniferous forests are gradually becoming converted to mixed stands more adapted to drought (Temperli et al. 2012). European beech is the dominant tree species in these mixtures from the lowland to montane vegetation zones in Central Europe, playing a critical role in the transition and adaptation of forests to climate change (Tarp et al. 2000). The combinations of beech together with Norway spruce, oaks, Scots pine or maples form a perspective mixture, enhancing overproduction and resistance to drought periods (Pretzsch et al. 2013, 2021; Vacek et al. 2021). In light of climate change and its associated phenomena, advantage of beech mixture is that high increment and shoot development in beech occur earlier and are particularly compressed within the growing season. Consequently, the potential occurrence of drought during the current year, which frequently arises in the middle part of the growing season, is less likely to hinder these developments in beech as compared to spruce (Konôpka et al. 2014).

On the flip side, transitioning from spruce stands to beech stands has the potential to alter specific production-ecological processes, particularly those related to the carbon regime, within forest ecosystems. The alteration of the carbon cycle impacts forest ecosystem functioning by influencing the scale and timing of various carbon fluxes. It is imperative to thoroughly investigate these ecosystem changes before undertaking significant shifts in species composition toward beech dominance (Konôpka et al. 2013). In the future, beech may occupy a larger coverage in the lower hills and lower mountains, while its area in the lowlands and uplands may be reduced due to increased competitive pressure from oak (Silva et al. 2012).

The annual gross production of beech is primarily influenced by spring temperatures and, irregularly, by summer water stress (Chiesi et al. 2016). The growth response to climate change mainly shows in the long term (Büntgen et al. 2007) and through shifting in altitudinal vegetation zones (Vanoni et al. 2016). Understanding the ongoing shifts in vegetation zones is essential for the development of sustainable forest management strategies in the face of climate change (Kulhavý 2004). Altitudinal vegetation zones represent regional frameworks of climatic conditions characterizing the growth of forest tree species in commercial forests (Svobodová et al. 2009), which dominate temperate forests in Europe. Modeling climate change can provide support for the development of adaptation strategies for European beech (Mindáš & Škvarenina 1996) in forest stands (Fig. 3). It also provides a scientifically sound ecological and eco-

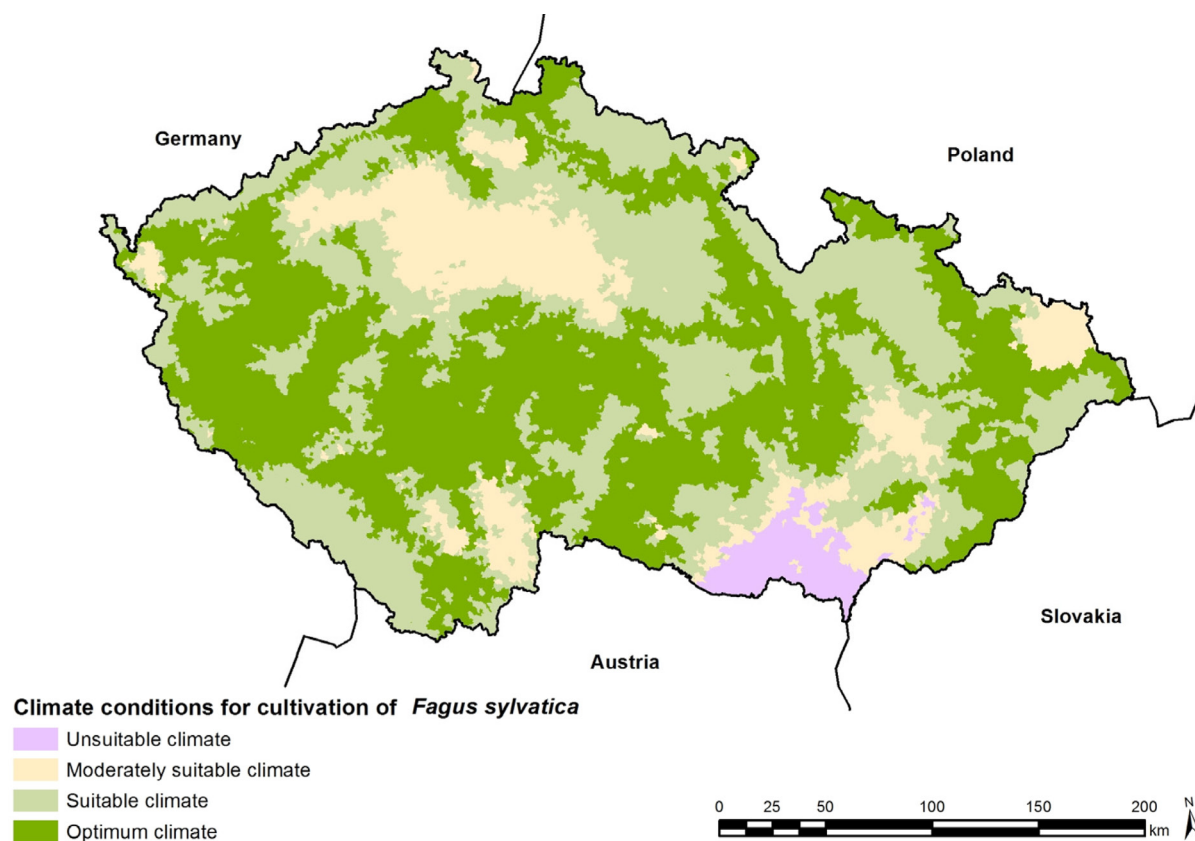


Fig. 3. Climatic conditions for growing European beech in the Czech Republic in 2050 (according to Machar et al. 2017), purple – unsuitable conditions, ochre – partially suitable conditions, light green – suitable conditions, dark green – optimal conditions.

nomic justification for forest management based on an adaptive approach to forest ecosystems (Machar et al. 2017).

6. Close-to-nature silvicultural management in the context of climate change

The survival and prosperity of European beech play a critical role in sustainable forest management, which not only includes its ecological performance but also its commercial significance (Shahverdi et al. 2013). In relation to global climate change, beech, especially in the center of its range, has been a highly successful tree species (Machar et al. 2017). Gessler et al. (2007) discuss opposite slopes and indicate that the northeastern slope is comparatively cold and wet, and this is where most current beech forests in Central Europe are. The southwestern slope is dry and warm, representing the potential climate for beech stands in Central Europe in the future (Kolář et al. 2017).

Close-to-nature management and its strategies should address both ecological and economic risks. Adaptation to climate change can lead to genetically and phenotypically structured populations throughout

the species' range within just a few generations, depending on the forest management system used (Brank et al. 2014). Increased warming and reduced precipitation diminish the increment and production of beech, especially at lower altitudes. With lower soil water content, nitrate uptake is also reduced. Significant nitrogen loss affects proper seedling growth, most often on calcareous substrates (Dannenmann et al. 2016). Reduced water availability and increased drought stress result in lower biomass production and higher mortality. Solar activity is also a substantial factor given climate change. The number and intensity of sunspots are crucial when assessing precipitation and seasonal temperature (Rötzer et al. 2013). From a silvicultural perspective, it is essential to exploit the natural environment for beech forest growth or to monitor the possible shift of altitudinal vegetation zones and adjust adaptive management accordingly: in particular, by using careful forest management strategies, promoting natural regeneration, and maximizing species mixing and differentiation of beech stands (species richness, balance, and diversity) in terms of higher production potential, resistance, and adaptation to climate change (Bošela et al. 2016; Vacek et al. 2021, 2023; Parhizkar et al. 2022).

If we want to apply the principles of close-to-nature management in beech stands, the first step is to change

the structure of the forest from age classes to diameter classes, which can primarily be achieved by systematic tending from the earliest stages (Štefančík et al. 2018b). Former methods of stand tending led to the less favorable (height-leveled) structure of beech stands. Therefore, it is necessary to build differentiated height structures through positive selection from above (Štefančík 2022). Moreover, this thinning method has a positive effect not only on increasing the value production but also mitigating the impact of climate change (Remeš et al. 2015). Specifically, Štefančík's free crown thinning (Štefančík 1984), aimed at cultivating so-called target trees (Štefančík 1984), has proven to be a suitable method (Fig. 4; Štefančík et al. 2014). After reaching their target diameter, gradual felling creates space and conditions for subsequent natural regeneration, which responds relatively quickly to gaps in the canopy (Šumichrast et al. 2023).

Therefore, in terms of progressive climate change, it is advisable to gradually shift from age-class forests to structurally differentiated, mosaic-structured forests with greater adaptability (Puettmann 2011; Messier et al. 2015; Puettmann et al. 2015; Bošela et al. 2016; Saniga et al. 2022). The shelterwood management method using selection principles best suits this purpose (Fig. 5; Poleno et al. 2009). The gradual reconstruction of these forests into a mosaic structure must start in a middle-aged stand, favoring local group-based natural regeneration within lengthier regeneration periods. This process will result in the formation of an abundance of regeneration elements at different stages of development, establishing structural differentiation of stands – both spatial (horizontal and vertical) and in age (Poleno et al. 2009). This cyclical pattern is characterized by the continuous dynamics of natural regeneration and spatial differentiation of stands (Schütz 1990; Saniga et al. 2022). As a shade-loving tree species, European beech is well-suited to form this mosaic stand structure (Saniga 2019). The small developmental cycle of close-to-nature beech forests often

creates a mosaic structure after small-scale disturbances (Mataji & Vahedi 2021). This is evident, for example, in the Jizera Mountains Beech Forest National Nature Reserve (Vacek et al. 2007), in Bažinky – Zone I of the Krkonoše National Park (Vacek et al. 1987; 2019), or in beech forests in Slovakia (Korpel 1985). Stand gaps and subsequent natural regeneration occur through the death of one or a few trees. For example, Kucbel et al. (2012) and Jaloviar et al. (2020) report that these gaps form a fairly even temporal pattern, which is the basis for the structural differentiation of these stands. Therefore, the silvicultural system of mosaic formation through group-selection management is based on these natural small-scale cyclical dynamics of close-to-nature beech stands. This involves a local group selection of trees in different diameter classes, especially those that have reached roughly the target diameter. After harvesting these trees, the conditions are created for natural regeneration in an area of a clump or group, at the most (Schütz 1990; Poleno et al. 2009; Vacek et al. 2015). In beech stands, the harvested area should not exceed 4–5 ares in size (Schütz 1990). According to Schütz (1990), European beech is a sympodial-branching tree, and therefore, requires individuals of the same height side by side in a clump to group arrangement in the youngest growth stages. In the upper layer, beech tends to enlarge its crown and form wolf trees, which reduces height growth (Poleno et al. 2009).

According to Hessenmöller et al. (2018), the stand volume of group-selected or mosaic beech stands should be in the range of 296–388 m³ ha⁻¹ at a target diameter of 66–70 cm, with a current annual increment of 6.7–7.8 m³ ha⁻¹. In selection forests on production-rich sites, Saniga (1996) considers an optimum stand volume of 340 m³ ha⁻¹ with a target diameter of 54 cm. According to Schaeffer & Schaeffer (1951), this is a stand volume of 200 to 250 m³ ha⁻¹, which is 30% lower than in age-class beech stands, but the production quality is generally higher. For comparison, Table 1 shows the basic production parameters of mature beech stands across Europe,

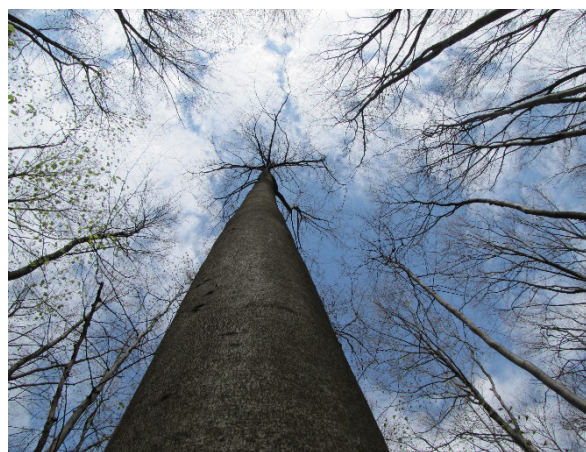


Fig. 4. Target beech at the age of 80 years with dbh = 47 cm after 40 years of Štefančík's free crown thinning in Central Slovakia (Photo: Igor Štefančík).

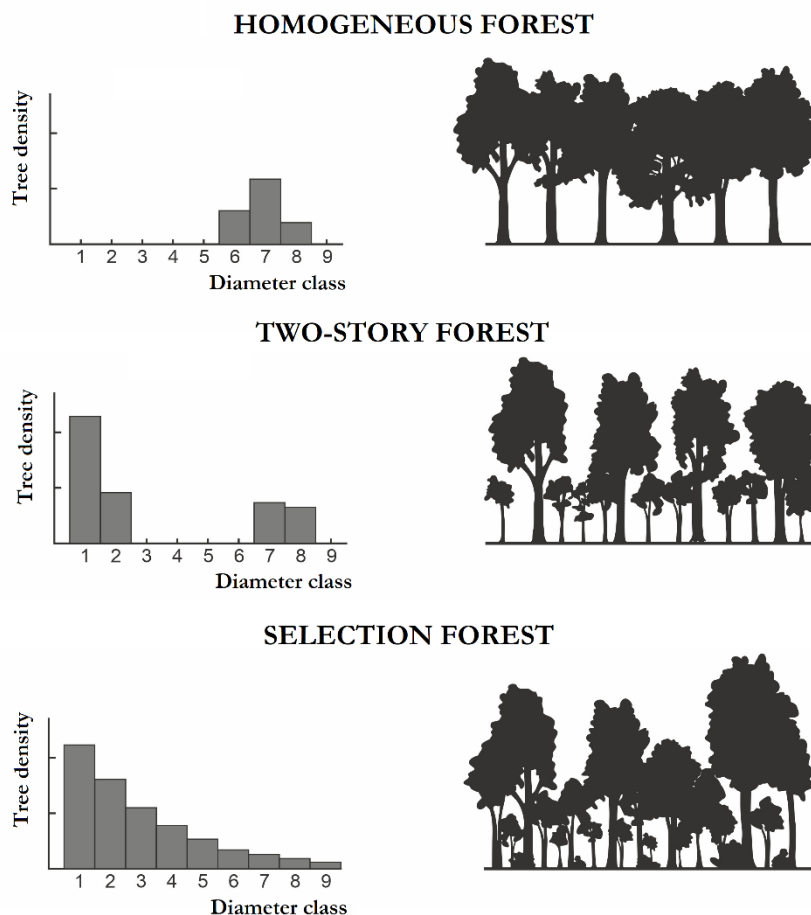


Fig. 5. Basic methods of beech cultivation, including the diameter structure of the stand (homogeneous forest – clear-cutting method, two-story forest – shelterwood method, fully differentiated forest – selection method).

with the average stand volume reaching $591 \text{ m}^3 \text{ ha}^{-1}$ and tree density of $317 \text{ trees ha}^{-1}$. Maximum stand volume was observed in the relic virgin beech forest Uholka in Ukrainian Carpathians – $1,271 \text{ m}^3 \text{ ha}^{-1}$ (Trotsiuk et al. 2012). The magnitude of the production quantity is determined by the required crown size in a group selection or mosaic forest, which is considerably larger (wider and longer) than in an age-class forest. Thus, larger crowns produce more biomass in the same period of time, i.e., they need less time to reach the target girth (Knoke 2002). Moreover, Račko et al. (2011) indicate that in beech trees with crown lengths greater than half the height of trees in the upper layer of the selection forest, the incidence of false heartwood is very low, i.e., a high production quality is expected.

The mosaic structure of broadleaved stands, especially beech, is a result of the concept of close-to-nature silviculture with significant adaptability to advancing climate change (Brang et al. 2014). Under ongoing climate change, mosaic beech stands achieve greater ecological stability and adaptability when compared to age-class forests (Lapin et al. 2008). This is due not only to their free thinning from above (Štefančík 1984, 2015), followed by spatial differentiation (Bošela et al. 2016) but also to a larger assimilative apparatus more resistant to

climate change consequences (Vacek 1987, 1988; Vacek et al. 2023). This is not a formation of selection forests, for which soil and climatic conditions are often unsuitable, but mosaic beech stands with a clump to group size, largely of up to 2 ares in size.

7. Conclusion

European beech, due to its relative resistance to abiotic and biotic factors, can play in future a major role in the tree species composition of Central European forests compared to dying stands of Norway spruce. Due to its character and shade-loving and euryvalent nature, this tree species has great potential to fulfill both production and non-production forest functions. In relation to climate change, the spread of beech to higher altitudes in suitable habitats should be approached with caution. Adaptive management favors creating structurally differentiated mosaic stands with European beech. In the context of close-to-nature silviculture, it is essential to promote structural differentiation of stands and the cultivation of mixed forests to achieve not only higher production potential but also to increase the resistance

Table 1. Overview of selected publications of dominant European beech forests related to production parameters in Europe.

Study	Country	Altitude [m a.s.l.]	Age [years]	DBH [cm]	Height [m]	Basal area [m ² ha ⁻¹]	Volume [m ³ ha ⁻¹]	Density [trees ha ⁻¹]
Ambrož et al. 2015	Czechia	432–451	175–188	57.5–59.0	30.4–39.0	39.2–50.3	732–975	152–184
Bílek et al. 2011	Czechia	345–501	159–189	35.9–61.7	26.4–41.2	27.2–40.3	506–864	93–272
Commarmot et al. 2005	Ukraine	700–800		48.0		38.5	770	219
Commarmot et al. 2005	Switzerland	600–700	150	41.6		30.7	524	259
Drössler & Lüpke 2007	Slovakia	500–820		35.4–38.8		32.5–35.0	451–716	290–320
Fuchs et al. 2021	Czechia	633–804	130–171	41–58	26–28		64–396	
Fuchs et al. 2021	Germany	662–665	149	43	27		247	
Hájek et al. 2021	Czechia	590–700	105–151	36.0–49.6	21.8–33.1	41.4–50.5	557–814	260–480
Jaworski & Kołodziej 2002	Poland	720–810				33.0–35.0	577–610	158–288
Kucbel et al. 2012	Slovakia	575–1,180		42.5–43.5		33.4–46.9	452–744	226–401
Matović et al. 2018	Serbia	400–1,175	max. 400	30.2–61.1	28.0–39.2	21.5–39.3	290–760	123–301
Petrian et al. 2012	Romania	350–620		50.3–52.2	33.1–34.2	37.3–38.3	650–676	487–499
Šimůnek et al. 2019	Czechia	740–790	149–182	41–49	26–31	34.3–36.5	347–458	154–256
Šimůnek et al. 2021	Poland	510	90–123	39	23		380	
Šimůnek et al. 2021	Italy	1,275–1,340	94–247	52–60	29–33		560–720	
Štefančík et al. 2018	Slovakia	510–700	83–105	37.7–44.3	32.4–34.1	36.9–46.8	561–706	375–770
Trotsiuk et al. 2012	Ukraine	660–730	max. 451			40.8–36.5	524–1,237	270–590
Vacek et al. 2019a	Czechia	420–440	104–158	35.0–44.2	16.8–26.8	36.7–47.0	591–699	240–488
Vacek et al. 2019b	Czechia	810	168	43.7	30.0	48.0	710	320
Vacek et al. 2021	Czechia	980–1,065	234	41.2–44.7	21.9–24.6	30.5–41.0	365–445	236–261
Vančura et al. 2022	Czechia	412	99	37.7	27.9	41.4	590	372
von Oheimb et al. 2005	Germany	100		29		32.7	605	263

Note: DBH – diameter at breast height.

and resilience of forest stands. At higher altitudes, beech and spruce represent an ideal type of mixture, as it is a combination of deciduous and coniferous trees with different ecological requirements and different morphology of the root system. In the lowlands, the admixture of beech in oak and Scots pine stands creates a mixture that maintains overproduction even in periods of drought. A mixture of Norway maple and sycamore maple also appears to be resistant to drought in beech stands. If it is planned to establish a beech stand in connection with another shade-loving tree, such as silver fir, it is important to consider the ecology of these tree species. However, the advancement of natural regeneration and the subsequent growth of low stories is not possible without the harmonization of forest ecosystems with regard to the abundance of ungulate, which can locally limit natural regeneration. In conclusion, foresters must be aware of the spread of new threats in future, such as beech bark beetles, as a result of climate change.

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